RAUNAK DEY

IIT Madras
B.S. Degree

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SUMMARY

Data Science and Machine Learning student with experience in AI systems, information retrieval, quantitative research, and full-stack engineering. Builds and evaluates data-driven systems using machine learning, embeddings, retrieval pipelines, and statistical experimentation, with an interest in reliable AI and cloud-based systems.

EDUCATION

Indian Institute of Technology Madras — Chennai, Tamil Nadu
BS in Data Science and Applications | CGPA: 7.21/10 — *September 2022 – August 2027*

SKILLS

Languages: Python, SQL, TypeScript, JavaScript
Frameworks: FastAPI, Next.js, React, Flask
AI/ML: PyTorch, Scikit-learn, Sentence Transformers
Data & Infrastructure: PostgreSQL, pgvector, Redis, Docker

EXPERIENCE

Quantitative Research Consultant — April 2024 – March 2025
WorldQuant — *Remote*

- Developed and evaluated 15 quantitative alphas across price-volume, news, options, and social-media datasets, using 25 unique operators and 28 financial data fields
- Completed 4 alpha pyramids spanning News, Options, Price-Volume, and Social Media datasets
- Achieved combined alpha performance of 0.82 and Power Pool performance of 1.12 through iterative signal research and evaluation

PROJECTS

Mind Palace | *Python, FastAPI, PostgreSQL, pgvector, MCP* — August 2026 – Present

- Built open-source AI memory infrastructure that transforms knowledge corpora into attributable, model-ready context through incremental ingestion, embeddings, hybrid retrieval, and token-budgeted context packing
- Implemented vector, keyword, and Reciprocal Rank Fusion (RRF) retrieval using pgvector and pg_trgm, achieving 87% Recall@5 at $\sim$51ms latency on a 202-document benchmark
- Designed an evaluation and quality-gating pipeline over 98 queries across 14 categories using Recall@K, MRR, nDCG, and paired bootstrap confidence intervals to detect retrieval regressions

Finalysis 2.0 | *Next.js 16, TypeScript, Redis, Tailwind CSS* — May 2025 – June 2026

- Built an investor-facing stock research platform covering 2,300+ NSE equities with fundamental metrics, price momentum, news sentiment, and explainable recommendations
- Engineered a resilient multi-source data pipeline integrating Yahoo Finance, Screener.in, Google News RSS, Redis caching, and daily market snapshots with source fallbacks and freshness-aware data handling
- Engineered fault-tolerant serverless APIs with rate limiting, circuit breakers, cache fallbacks, retries, and automated price-refresh workflows, deployed at fin-alysis.vercel.app

CV-Embed | *React 19, TypeScript, Vite, @react-pdf/renderer* — February 2026 – July 2026

- Built an ATS-friendly resume builder with real-time preview, structured editing, browser persistence, and configurable templates, deployed at cv-embed.vercel.app
- Engineered PDF, DOCX, and normalized JSON export pipelines with Zod-based validation and automated resume quality scoring
- Developed an embeddable SDK v2 using a postMessage event API with auto-height iframe integration, host controls, and lifecycle callbacks for external portals

ACCOMPLISHMENTS

Kaggle August 2025 – Present

Notebooks Master | Rank 75/60,358

- Earned **3 Gold, 8 Silver, and 8 Bronze medals** across 19 completed competitions, including 22 solo competition entries
- **BirdCLEF+ 2026:** Ranked 795 out of 4,094 teams by developing an ensemble pipeline combining ProtoSSM and BirdNET models
- **NVIDIA Nemotron Model Reasoning Challenge:** Ranked 1,715 out of 4,182 teams in an LLM open-model reasoning benchmark

International Quant Championship 2024 March 2024 – June 2024

WorldQuant

- Ranked 165 globally in Stage 1 and 184 globally in Stage 2 among thousands of participants
- Contributed to a team OS Score of 36,000+ using 24+ financial data fields for quantitative signal generation
- Collaborated in a two-member team to design and implement alpha strategies under competition constraints

CERTIFICATIONS

OCI Certified Foundations Associate
Credential ID: 331754025OCI26FNDCFA

OCI Certified AI Foundations Associate
Credential ID: 331754025OCI26AICFA

EXTRA-CURRICULAR ACTIVITIES

Deputy Secretary March 2024 – August 2025
Diplomacia, the IITM Soft Skills Society *SA Contribution Certificate*

Marketing Team Volunteer August 2024 – December 2024
Student Placement Council, IITM BS Degree *SPC Impact Report*